Makayla Choudary

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Education

New York University — New York, New York
Bachelor of Science in Computer Engineering — *Expected Graduation: May 2027*

- Relevant Coursework: Data Structures, Computer Architecture, Machine Learning, Digital Logic, Circuits, Linear Algebra, Statistics

Experience

Data Analytics Intern — Summer 2025
Travel Resorts of America — *Remote / North Carolina*

- Built and automated Power BI dashboards tracking occupancy, cancellations, and site utilization across multiple resort properties, improving visibility into operational performance.
- Designed Excel Power Query transformations and DAX measures to standardize key operational metrics, including unused sites and active reservations.
- Developed Python and Databricks workflows using Snowflake datasets to analyze loan delinquency milestones across 30/60/90/120+ day thresholds.
- Implemented Collibra Data Quality rules to validate completeness and identify discrepancies across booking, financial, and site-level data sources.
- Delivered executive-facing performance reports used to support operational planning, reporting accuracy, and management decision-making.

Projects

Predictive Maintenance ML System | *Python, pandas, scikit-learn, Jupyter, Git/GitHub* — Spring 2026

- Built a machine learning pipeline to predict equipment failure risk from industrial sensor data using temperature, torque, rotational speed, and tool wear features.
- Addressed severe class imbalance, with failures representing 3.39% of samples, by prioritizing recall and F2 score over accuracy.
- Compared Logistic Regression, Random Forest, and Gradient Boosting models, applying threshold optimization to prioritize failure detection.
- Evaluated models using confusion matrices, recall, F2 score, and feature importance to compare failure detection performance.

LSTM Time-Series Prediction Project | *Python, pandas, scikit-learn, matplotlib, Jupyter, Git* — Fall 2025

- Built and trained an LSTM neural network to model sequential time-series data and forecast future values.
- Created preprocessing workflows for data cleaning, normalization, train-test splitting, and sequence generation.
- Evaluated model performance using prediction plots and error metrics to compare predicted values against actual trends.
- Experimented with architecture depth, sequence length, and hyperparameters to improve forecasting accuracy.

Robot Design Project | *Raspberry Pi, Python* — Fall 2025

- Built a Raspberry Pi–powered autonomous robot to solve a 15-tile puzzle using motor control logic and movement sequencing.
- Integrated motor wiring, GPIO control, and Python timing logic to coordinate reliable robot movement under competition constraints.
- Placed Top 6 out of 50 teams in a course-wide engineering design competition.

Technical Skills

Languages: Python, C++, Java, SQL
Data/ML: pandas, NumPy, scikit-learn, matplotlib, Power BI, Excel Power Query, DAX
Cloud/Data Tools: Databricks, Snowflake, Collibra Data Quality
Hardware/Engineering: Raspberry Pi, Arduino, GPIO, motor control, digital logic, circuits
Developer Tools: Git, GitHub, Jupyter Notebook